

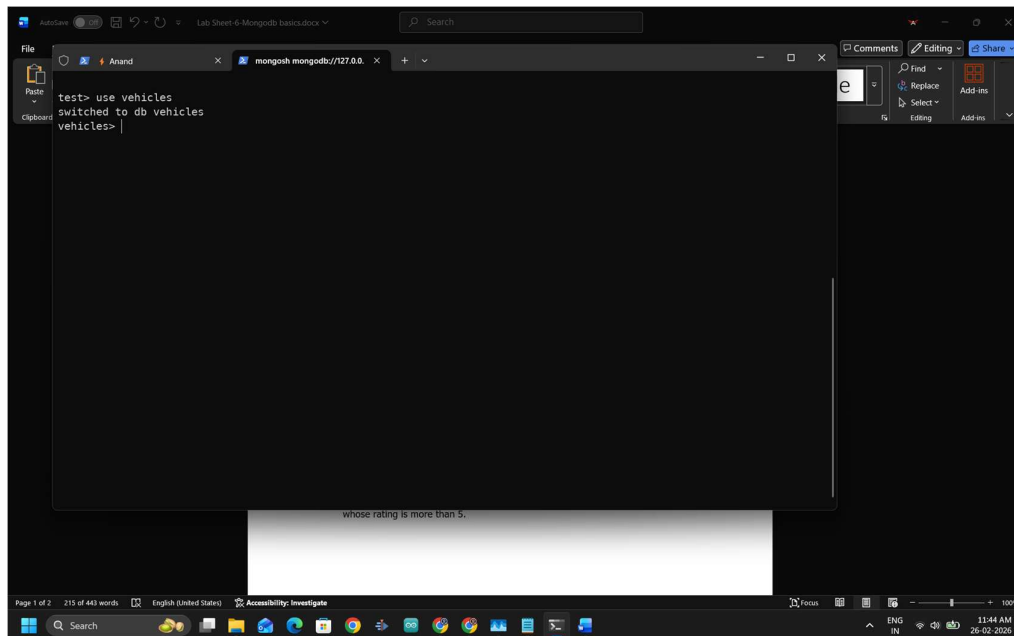
Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech
Faculty Name: Prof. S.Gopikrishnan
Student name: Anand Kumar V

Date: 26-Feb-26
School: SCOPE
Reg. no.: 23BCE8757

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".

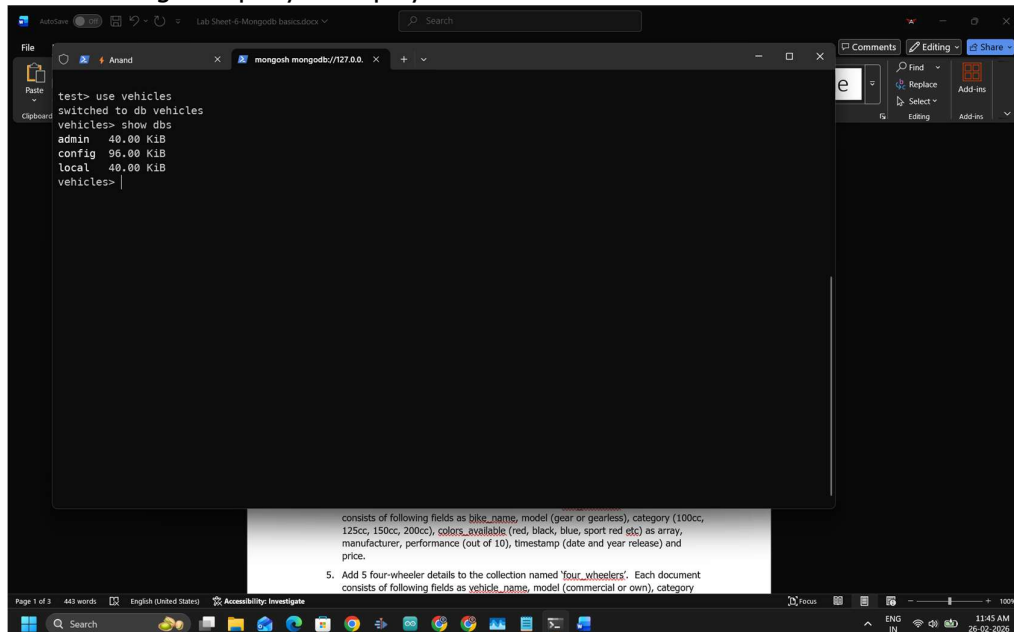


The screenshot shows a terminal window with the following commands and output:

```
test> use vehicles
switched to db vehicles
vehicles> |
```

The terminal window is titled 'mongosh mongodb://127.0.0.1'. The output of the 'use vehicles' command is 'switched to db vehicles'. The prompt 'vehicles>' is shown, indicating the database has been successfully selected.

2. Write a MongoDB query to display all the databases.



The screenshot shows a terminal window with the following commands and output:

```
test> use vehicles
switched to db vehicles
vehicles> show dbs
admin    40.00 KiB
config  96.00 KiB
local   40.00 KiB
vehicles> |
```

The terminal window is titled 'mongosh mongodb://127.0.0.1'. The output of the 'show dbs' command is a list of databases and their sizes: 'admin 40.00 KiB', 'config 96.00 KiB', and 'local 40.00 KiB'. The prompt 'vehicles>' is shown, indicating the database has been successfully selected.

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

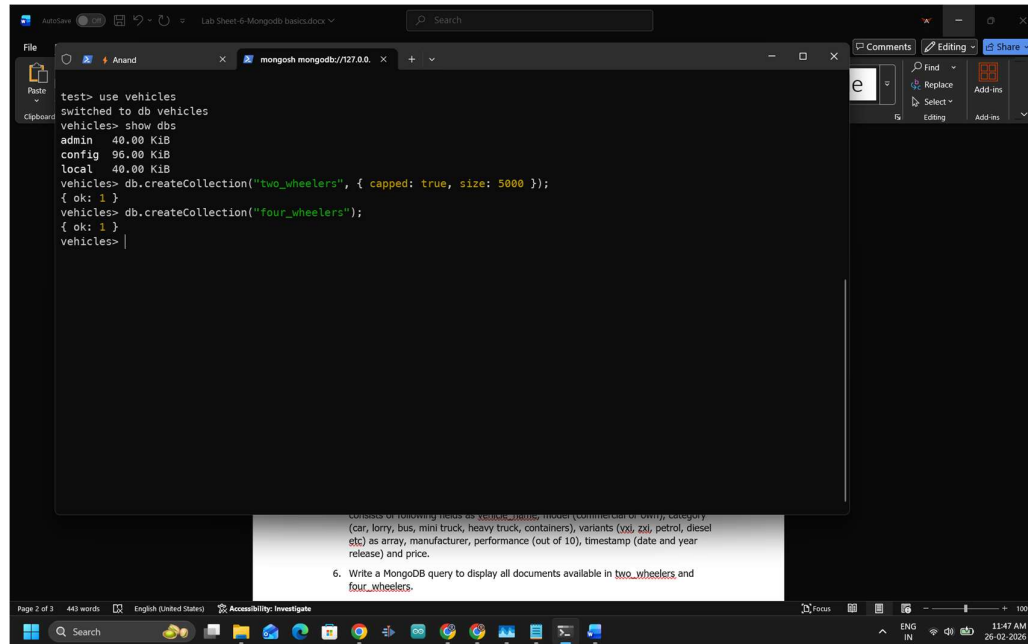
Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.



```
test> use vehicles
switched to db vehicles
vehicles> show dbs
admin    40.00 KiB
config  96.00 KiB
local   40.00 KiB
vehicles> db.createCollection("two_wheelers", { capped: true, size: 5000 });
{ ok: 1 }
vehicles> db.createCollection("four_wheelers");
{ ok: 1 }
vehicles> |
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

price.

```

vehicles> db.two_wheelers.insertMany([
  { bike_name: "PulsarNS200", model: "gear", category: "199cc", colors_available: ["black", "red"], manufacturer: "Bajaj", performance: 5, timestamp: new Date("2023-03-03"), price: 134000 },
  { bike_name: "Activa6G", model: "gearless", category: "125cc", colors_available: ["white", "black", "red"], manufacturer: "Honda", performance: 3, timestamp: new Date("2023-05-12"), price: 85000 },
  { bike_name: "ApacheRTR310", model: "gear", category: "312cc", colors_available: ["black", "sport red", "yellow"], manufacturer: "TVS", performance: 7, timestamp: new Date("2023-09-20"), price: 240000 },
  { bike_name: "1390SuperPulser", model: "gear", category: "1350cc", colors_available: ["orange", "black"], manufacturer: "KTM", performance: 10, timestamp: new Date("2024-02-15"), price: 2295000 },
  { bike_name: "SuperMeteor650", model: "gear", category: "648cc", colors_available: ["green", "grey", "black"], manufacturer: "RoyalEnfield", performance: 9, timestamp: new Date("2023-01-16"), price: 436000 }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fe970b1572bd67a7c298c'),
    '1': ObjectId('699fe970b1572bd67a7c298d'),
    '2': ObjectId('699fe970b1572bd67a7c298e'),
    '3': ObjectId('699fe970b1572bd67a7c298f'),
    '4': ObjectId('699fe970b1572bd67a7c2990')
  }
}
vehicles>
vehicles>
  
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (own, commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```

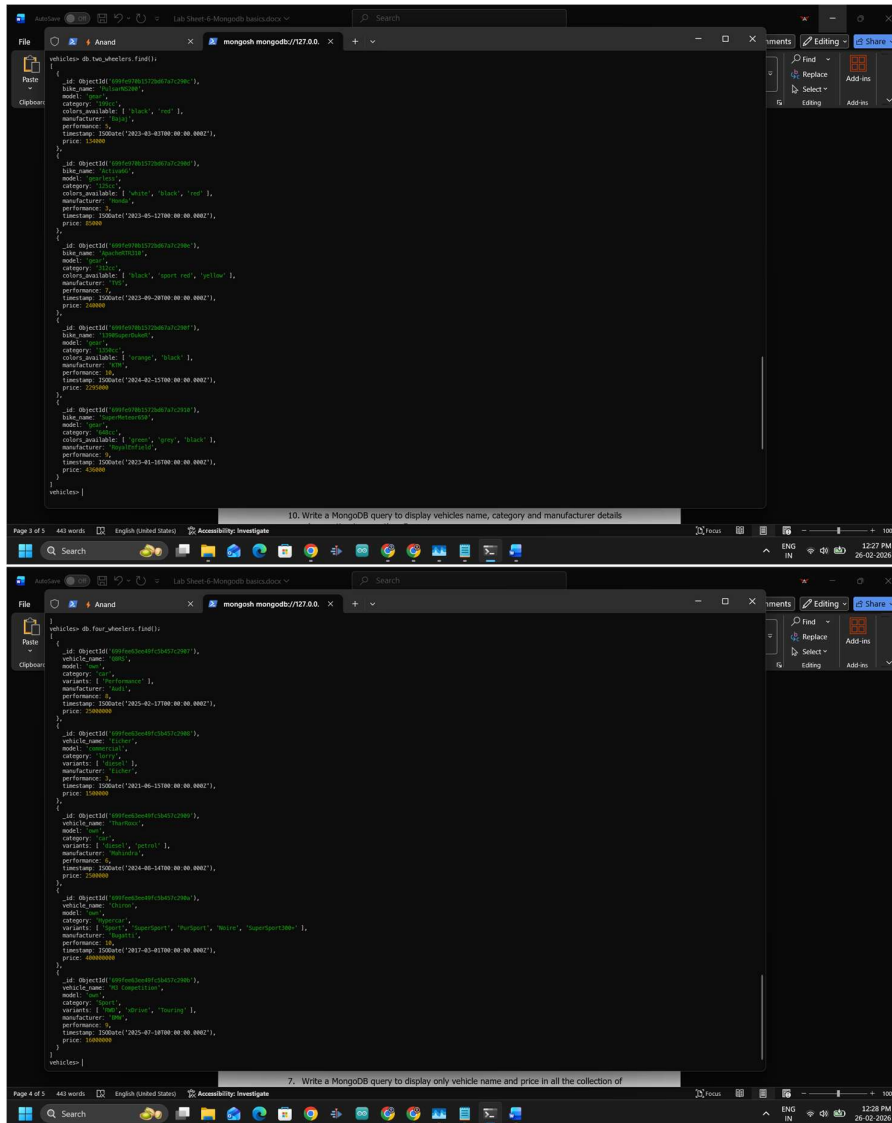
vehicles> db.four_wheelers.insertMany([
  { vehicle_name: "Swift", model: "own", category: "car", variants: ["vxi", "zxi", "petrol"], manufacturer: "Maruti", performance: 8, timestamp: new Date("2022-03-10"), price: 700000 },
  { vehicle_name: "Eicher", model: "commercial", category: "lorry", variants: ["diesel"], manufacturer: "Eicher", performance: 7, timestamp: new Date("2021-06-15"), price: 1500000 },
  { vehicle_name: "08R5", model: "own", category: "car", variants: ["Performance"], manufacturer: "Audi", performance: 8, timestamp: new Date("2025-02-17"), price: 25000000 },
  { vehicle_name: "Eicher", model: "commercial", category: "lorry", variants: ["diesel"], manufacturer: "Eicher", performance: 3, timestamp: new Date("2021-06-15"), price: 1500000 },
  { vehicle_name: "TharRox", model: "own", category: "car", variants: ["diesel", "petrol"], manufacturer: "Mahindra", performance: 6, timestamp: new Date("2024-08-14"), price: 2500000 },
  { vehicle_name: "Chiron", model: "own", category: "Hypercar", variants: ["Sport", "SuperSport", "PurSport", "Noire", "Supersport300s"], manufacturer: "Bugatti", performance: 10, timestamp: new Date("2017-03-01"), price: 40000000 },
  { vehicle_name: "M3 Competition", model: "own", category: "Sport", variants: ["RWD", "xDrive", "Touring"], manufacturer: "BMW", performance: 9, timestamp: new Date("2025-07-10"), price: 16000000 }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fee33ee49fcb457c2987'),
    '1': ObjectId('699fee33ee49fcb457c2988'),
    '2': ObjectId('699fee33ee49fcb457c2989'),
    '3': ObjectId('699fee33ee49fcb457c298a'),
    '4': ObjectId('699fee33ee49fcb457c298b')
  }
}
vehicles>
  
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech
Faculty Name: Prof. S.Gopikrishnan
Student name: Anand Kumar V

Date: 26-Feb-26
School: SCOPE
Reg. no.: 23BCE8757

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.



```
vehicles: db.two_wheelers.find();
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b1"),
  "bike_name": "YamahaYZF01R",
  "model": "YZF01R",
  "category": "bike",
  "colors_available": [ "black", "red" ],
  "manufacturer": "Yamaha",
  "performance": 5,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 150000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b2"),
  "bike_name": "KTM1190RC",
  "model": "RC1190",
  "category": "bike",
  "colors_available": [ "white", "black", "red" ],
  "manufacturer": "KTM",
  "performance": 5,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 80000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b3"),
  "bike_name": "DucatiPanigaleV4",
  "model": "PanigaleV4",
  "category": "bike",
  "colors_available": [ "black", "sport red", "yellow" ],
  "manufacturer": "Ducati",
  "performance": 7,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 200000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b4"),
  "bike_name": "BMWHP1000",
  "model": "HP1000",
  "category": "bike",
  "colors_available": [ "orange", "black" ],
  "manufacturer": "BMW",
  "performance": 10,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 200000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b5"),
  "bike_name": "Supermoto",
  "model": "Supermoto",
  "category": "bike",
  "colors_available": [ "green", "grey", "black" ],
  "manufacturer": "Supermoto",
  "performance": 5,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 40000
},
}
vehicles:

10. Write a MongoDB query to display vehicles name, category and manufacturer details
```

```
vehicles: db.four_wheelers.find();
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b6"),
  "vehicle_name": "BMW",
  "model": "BMW",
  "category": "car",
  "variants": [ "Performance" ],
  "manufacturer": "BMW",
  "performance": 6,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 2500000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b7"),
  "vehicle_name": "Audi",
  "model": "Audi",
  "category": "car",
  "variants": [ "Sport" ],
  "manufacturer": "Audi",
  "performance": 5,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 1500000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b8"),
  "vehicle_name": "Ferrari",
  "model": "Ferrari",
  "category": "car",
  "variants": [ "Sport", "petrol" ],
  "manufacturer": "Ferrari",
  "performance": 5,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 2500000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20b9"),
  "vehicle_name": "Porsche",
  "model": "Porsche",
  "category": "car",
  "variants": [ "Sport", "Petrol", "Hybrid", "Electric", "SuperSport", "Petrol", "Hybrid", "SuperSport1000" ],
  "manufacturer": "Porsche",
  "performance": 10,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 4000000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20ba"),
  "vehicle_name": "Mazda",
  "model": "Mazda",
  "category": "car",
  "variants": [ "Sport" ],
  "manufacturer": "Mazda",
  "performance": 5,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 1000000
},
{
  "_id": ObjectId("65f5b8b9f1c8b20a2c20bb"),
  "vehicle_name": "Mazda",
  "model": "Mazda",
  "category": "car",
  "variants": [ "Sport", "Petrol", "Hybrid" ],
  "manufacturer": "Mazda",
  "performance": 5,
  "timestamp": ISODate("2023-02-02T00:00:00.000Z"),
  "price": 1000000
},
}
vehicles:

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

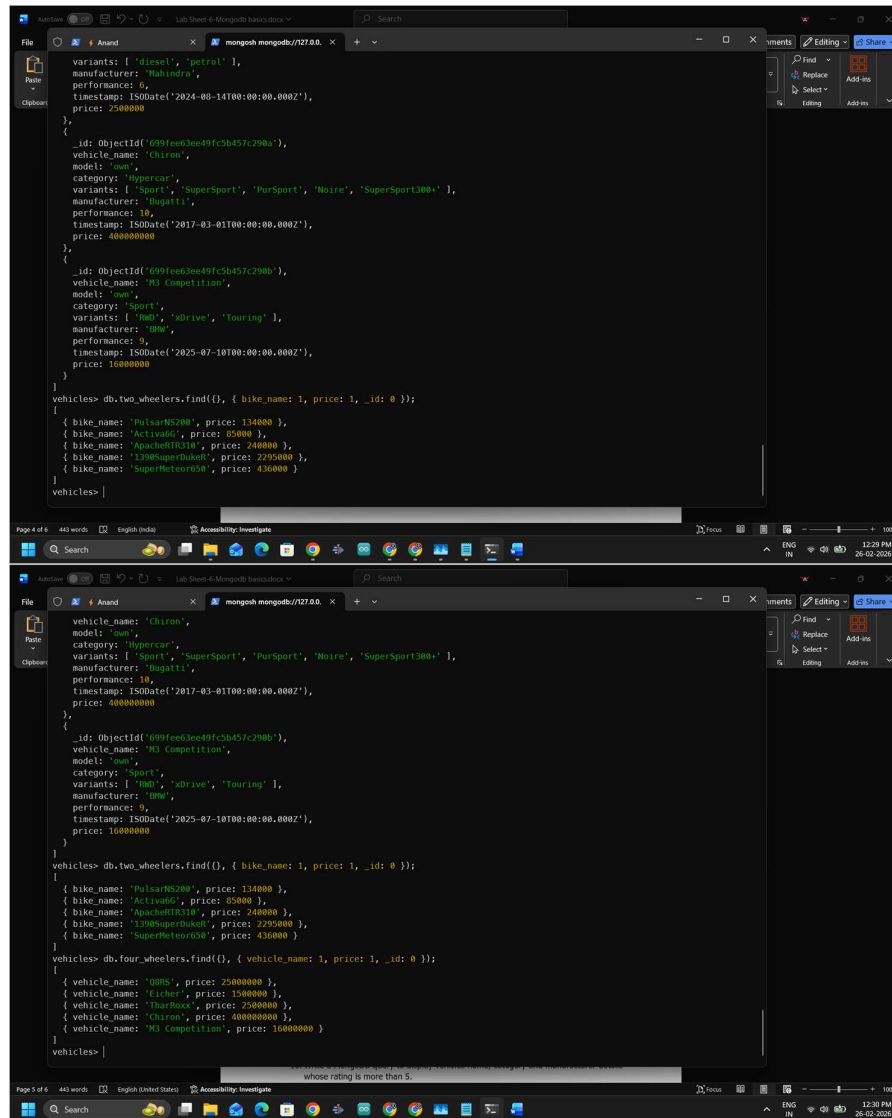
Date: 26-Feb-26

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757



The image shows a Windows 11 desktop with two Notepad++ windows open. The top window displays a JSON array of vehicle data, including details like _id, vehicle_name, model, category, variants, manufacturer, performance, timestamp, and price. The bottom window shows MongoDB queries and their results. The first query filters for two-wheelers, and the second query filters for four-wheelers. Both queries use the 'vehicles' database and the 'vehicles' collection. The results are displayed as JSON arrays.

```
variants: [ 'diesel', 'petrol' ],
manufacturer: 'Mahindra',
performance: 6,
timestamp: ISODate('2024-08-14T00:00:00.000Z'),
price: 2500000
},
{
  _id: ObjectId('699fee3ee49fc5b457c290a'),
  vehicle_name: 'Chiron',
  model: 'own',
  category: 'Hypercar',
  variants: [ 'Sport', 'SuperSport', 'PurSport', 'Noire', 'SuperSport300+' ],
  manufacturer: 'Bugatti',
  performance: 10,
  timestamp: ISODate('2017-03-01T00:00:00.000Z'),
  price: 400000000
},
{
  _id: ObjectId('699fee3ee49fc5b457c290b'),
  vehicle_name: 'R3 Competition',
  model: 'own',
  category: 'Sport',
  variants: [ 'RWD', '4Drive', 'Touring' ],
  manufacturer: 'BMW',
  performance: 9,
  timestamp: ISODate('2025-07-10T00:00:00.000Z'),
  price: 10000000
}
]
vehicles> db.two_wheelers.find( { bike_name: 1, price: 1, _id: 0 });
{
  bike_name: 'PulsarNS200', price: 134000 },
{
  bike_name: 'Activa6G', price: 85000 },
{
  bike_name: 'ApacheRTR10', price: 240000 },
{
  bike_name: '1390SuperDuke', price: 2295000 },
{
  bike_name: 'SuperMotor038', price: 436000 }
}
vehicles>
[
  {
    vehicle_name: 'OWN', price: 250000000 },
  {
    vehicle_name: 'Eicher', price: 15000000 },
  {
    vehicle_name: 'BharatBenz', price: 2900000 },
  {
    vehicle_name: 'Chiron', price: 400000000 },
  {
    vehicle_name: 'R3 Competition', price: 10000000 }
]
vehicles>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

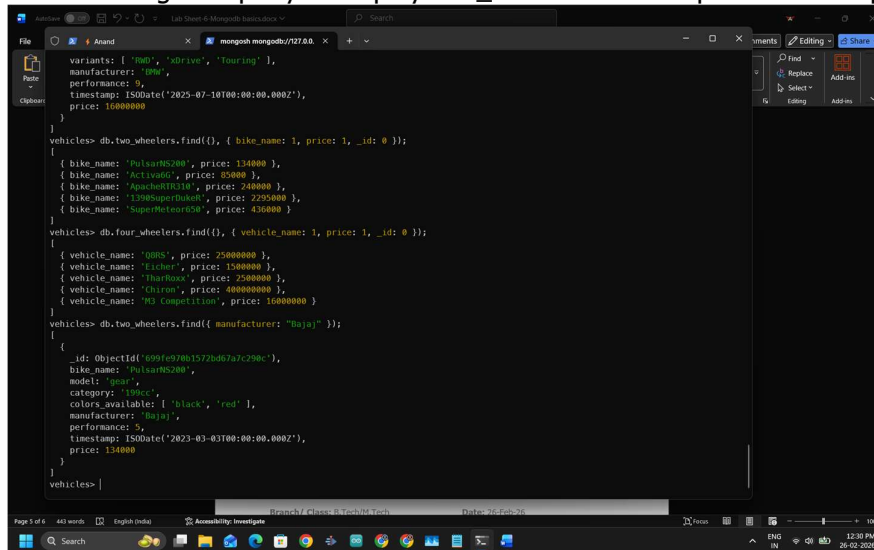
Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

8. Write a MongoDB query to display two_wheelers from a particular company



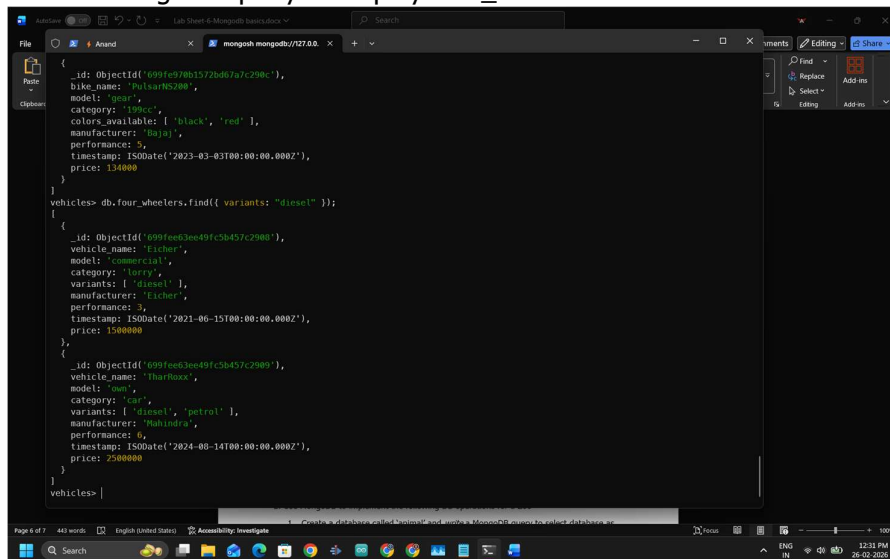
```
vehicles> db.two_wheelers.find( { bike_name: 1, price: 1, _id: 0 } );
{
  bike_name: 'PulsarNS200', price: 134000 },
{
  bike_name: 'Activ6G', price: 85000 },
{
  bike_name: 'ApacheR150', price: 240000 },
{
  bike_name: '139SuperDuke', price: 2295000 },
{
  bike_name: 'SuperMeteor650', price: 436000 }
}

vehicles> db.four_wheelers.find( { vehicle_name: 1, price: 1, _id: 0 } );
{
  vehicle_name: '200S', price: 25000000 },
{
  vehicle_name: 'Eicher', price: 1500000 },
{
  vehicle_name: 'TharRox', price: 2500000 },
{
  vehicle_name: 'Chiron', price: 400000000 },
{
  vehicle_name: '90 Competition', price: 16000000 }
}

vehicles> db.two_wheelers.find( { manufacturer: 'Bajaj' } );
{
  _id: ObjectId('699fe97061572b067a7c290c'),
  bike_name: 'PulsarNS200',
  model: 'gear',
  category: 'bikes',
  colors_available: [ 'black', 'red' ],
  manufacturer: 'Bajaj',
  performance: 5,
  timestamp: ISODate('2023-03-03T00:00:00.000Z'),
  price: 134000 }
}

vehicles>
```

9. Write a MongoDB query to display four_wheelers available in diesel variants



```
vehicles> db.four_wheelers.find( { variants: 'diesel' } );
{
  _id: ObjectId('699fe97061572b067a7c290c'),
  bike_name: 'PulsarNS200',
  model: 'gear',
  category: 'bikes',
  colors_available: [ 'black', 'red' ],
  manufacturer: 'Bajaj',
  performance: 5,
  timestamp: ISODate('2023-03-03T00:00:00.000Z'),
  price: 134000 }
}

vehicles> db.four_wheelers.find( { variants: 'diesel' } );
{
  _id: ObjectId('699fe97061572b067a7c290c'),
  vehicle_name: 'Eicher',
  model: 'commercial',
  category: 'lorry',
  variants: [ 'diesel' ],
  manufacturer: 'Eicher',
  performance: 3,
  timestamp: ISODate('2021-06-15T00:00:00.000Z'),
  price: 1500000 },
{
  _id: ObjectId('699fe97061572b067a7c290c'),
  vehicle_name: 'TharRox',
  model: 'own',
  category: 'car',
  variants: [ 'diesel', 'petrol' ],
  manufacturer: 'Mahindra',
  performance: 6,
  timestamp: ISODate('2024-08-14T00:00:00.000Z'),
  price: 2500000 }
}

vehicles>
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

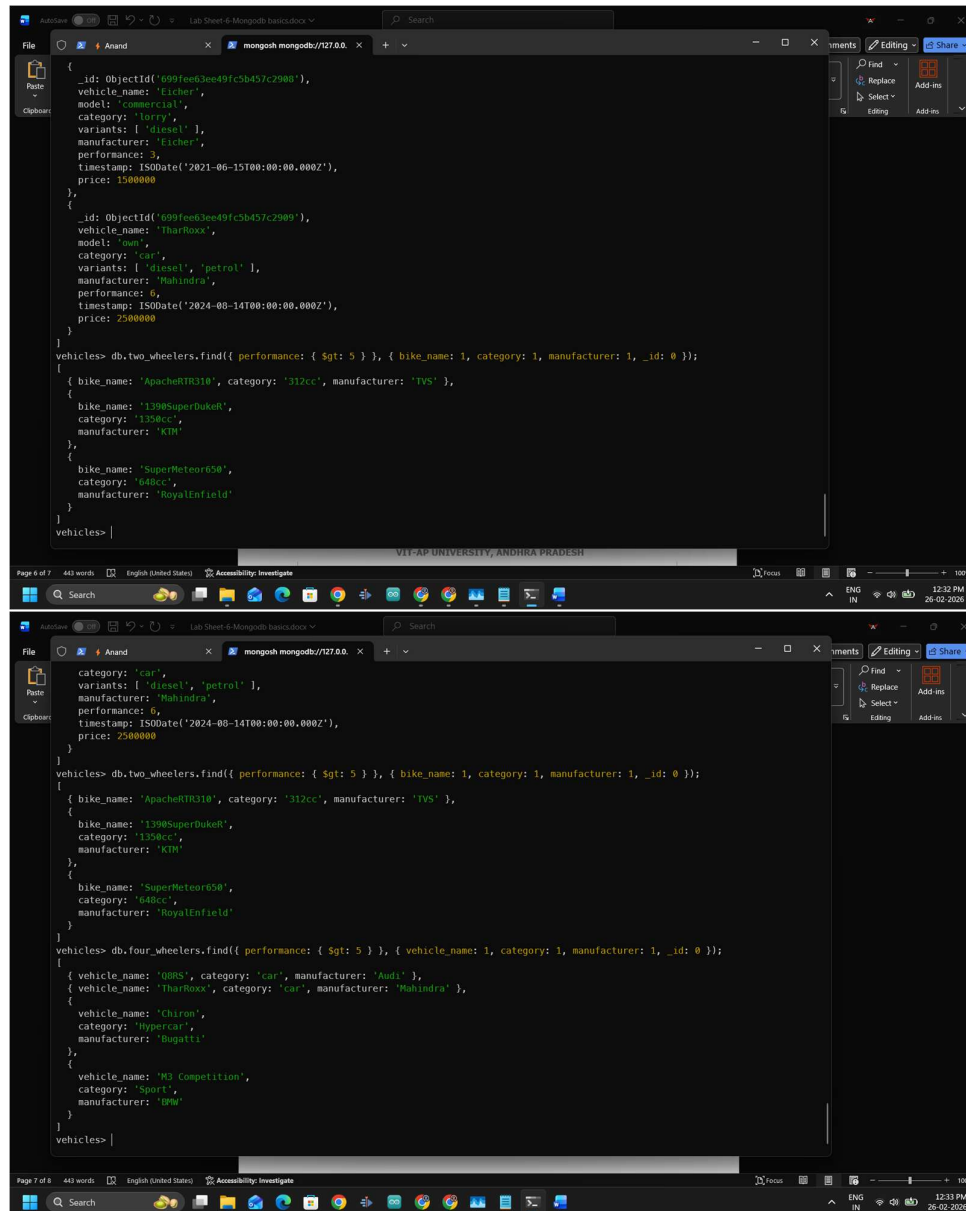
Date: 26-Feb-26

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757



The image displays two screenshots of a MongoDB shell interface, likely running in a web browser or a terminal window. The interface shows a collection of documents in a database, with fields such as _id, vehicle_name, model, category, variants, manufacturer, performance, timestamp, and price. The documents represent different vehicles, including a commercial lorry, a car, and a motorcycle. The shell prompt 'vehicles>' is visible, indicating the current database and collection context. The first screenshot shows a query that filters documents by performance (greater than or equal to 5) and returns the bike_name, category, and manufacturer. The second screenshot shows a similar query but with a different filter (performance greater than or equal to 5) and returns the vehicle_name, category, and manufacturer. The results are displayed as JSON documents.

```
{
  "_id": ObjectId("699feb3ee49fc5b457c2980"),
  "vehicle_name": "Eicher",
  "model": "commercial",
  "category": "lorry",
  "variants": [ "diesel" ],
  "manufacturer": "Eicher",
  "performance": 3,
  "timestamp": ISODate("2021-06-15T00:00:00.000Z"),
  "price": 1500000
},
{
  "_id": ObjectId("699feb3ee49fc5b457c2989"),
  "vehicle_name": "TharRoxa",
  "model": "van",
  "category": "car",
  "variants": [ "diesel", "petrol" ],
  "manufacturer": "Mahindra",
  "performance": 6,
  "timestamp": ISODate("2024-08-14T00:00:00.000Z"),
  "price": 2500000
}
]
vehicles> db.two_whealers.find( { performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, _id: 0 });
{
  "bike_name": "ApacheRTR310",
  "category": "312cc",
  "manufacturer": "TVS"
},
{
  "bike_name": "1390SuperDukeR",
  "category": "1350cc",
  "manufacturer": "KTM"
},
{
  "bike_name": "SuperMeteor650",
  "category": "648cc",
  "manufacturer": "RoyalEnfield"
}
]
vehicles>
{
  "category": "car",
  "variants": [ "diesel", "petrol" ],
  "manufacturer": "Mahindra",
  "performance": 6,
  "timestamp": ISODate("2024-08-14T00:00:00.000Z"),
  "price": 2500000
},
]
vehicles> db.two_whealers.find( { performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, _id: 0 });
{
  "bike_name": "ApacheRTR310",
  "category": "312cc",
  "manufacturer": "TVS"
},
{
  "bike_name": "1390SuperDukeR",
  "category": "1350cc",
  "manufacturer": "KTM"
},
{
  "bike_name": "SuperMeteor650",
  "category": "648cc",
  "manufacturer": "RoyalEnfield"
}
]
vehicles> db.four_whealers.find( { performance: { $gt: 5 } }, { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 });
{
  "vehicle_name": "Qoros",
  "category": "car",
  "manufacturer": "Audi"
},
{
  "vehicle_name": "TharRoxa",
  "category": "car",
  "manufacturer": "Mahindra"
},
{
  "vehicle_name": "Chiron",
  "category": "hypercar",
  "manufacturer": "Bugatti"
},
{
  "vehicle_name": "M3 Competition",
  "category": "sport",
  "manufacturer": "BMW"
}
]
vehicles>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

Faculty Name: Prof. S.Gopikrishnan

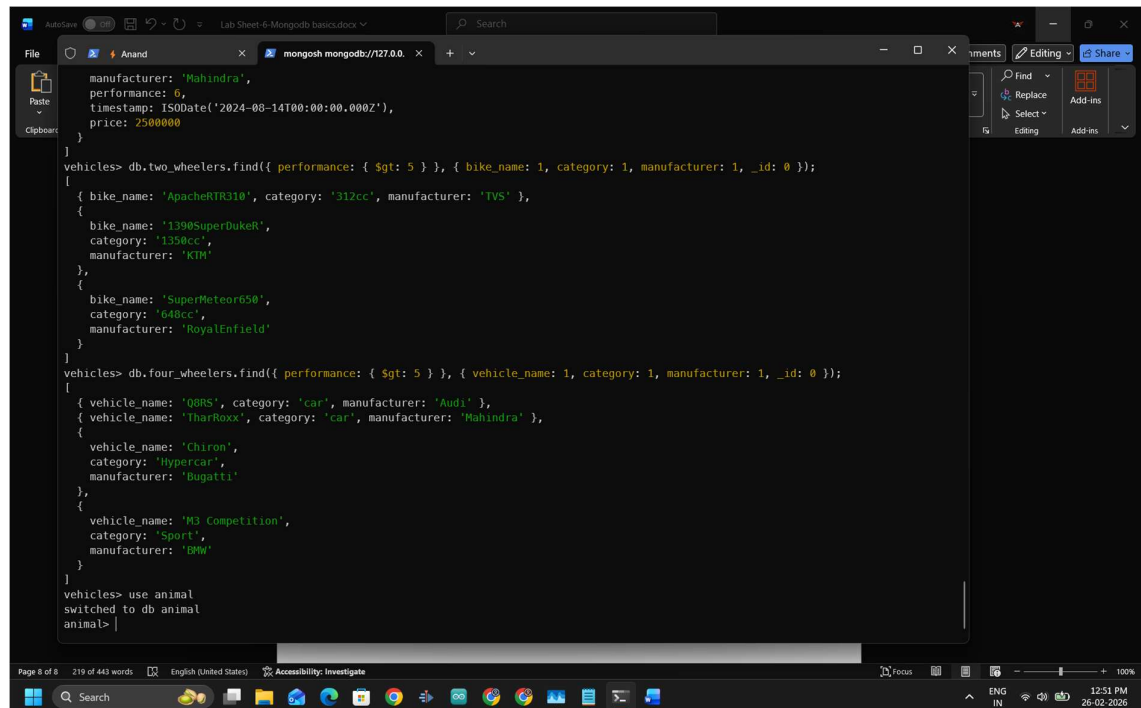
School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.



The screenshot shows a MongoDB shell window with the following content:

```
manufacter: 'Mahindra',
performance: 6,
timestamp: ISODate('2024-08-14T00:00:00.000Z'),
price: 2500000
}
vehicles> db.two_wheelers.find({ performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, _id: 0 });
{ bike_name: 'ApacheRTR310', category: '312cc', manufacturer: 'TVS' },
{
  bike_name: '1390SuperDukeR',
  category: '1350cc',
  manufacturer: 'KTM'
},
{
  bike_name: 'SuperMeteor650',
  category: '640cc',
  manufacturer: 'RoyalEnfield'
}
vehicles> db.four_wheelers.find({ performance: { $gt: 5 } }, { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 });
{ vehicle_name: 'Q8RS', category: 'car', manufacturer: 'Audi' },
{ vehicle_name: 'TharRoxx', category: 'car', manufacturer: 'Mahindra' },
{
  vehicle_name: 'Chiron',
  category: 'Hypercar',
  manufacturer: 'Bugatti'
},
{
  vehicle_name: 'M3 Competition',
  category: 'Sport',
  manufacturer: 'BMW'
}
vehicles> use animal
switched to db animal
animal>
```

The window also shows a file explorer on the left with a clipboard icon, and a right sidebar with search, replace, and add-ins options. The bottom status bar indicates 'Page 8 of 8', '219 of 443 words', 'English (United States)', 'Accessibility: Investigate', and system information like 'ENG IN', '12:51 PM', and '26-02-2026'.

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

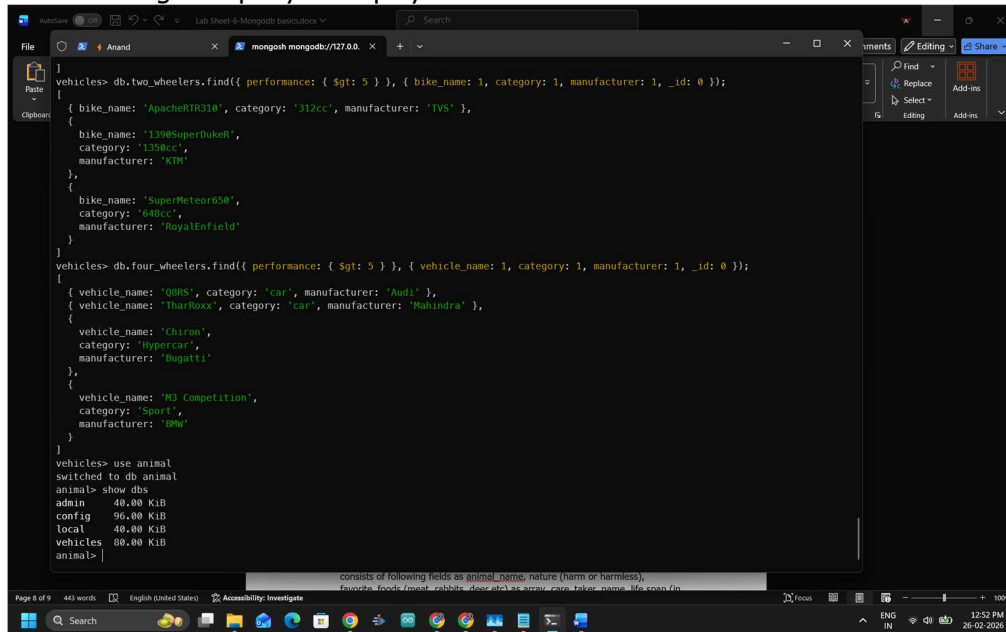
Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

2. Write a MongoDB query to display all the databases.



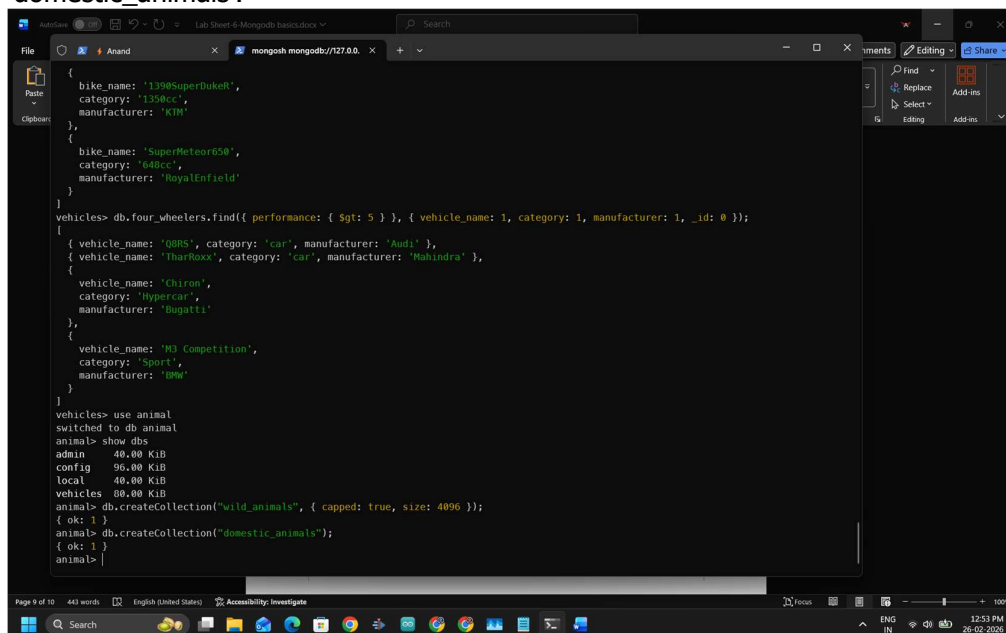
The screenshot shows a MongoDB terminal window with the following commands and output:

```
vehicles> db.wheelers.find({ performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, _id: 0 });
{
  bike_name: 'ApacheRTR310', category: '312cc', manufacturer: 'TVS' },
{
  bike_name: '1399SuperDukeR',
  category: '1358cc',
  manufacturer: 'KTM'
},
{
  bike_name: 'SuperMeteor650',
  category: '648cc',
  manufacturer: 'RoyalEnfield'
}
]

vehicles> db.four_wheelers.find({ performance: { $gt: 5 } }, { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 });
{
  vehicle_name: 'Q8RS', category: 'car', manufacturer: 'Audi' },
{
  vehicle_name: 'TharBoxx', category: 'car', manufacturer: 'Mahindra' },
{
  vehicle_name: 'Chiron',
  category: 'hypercar',
  manufacturer: 'Bugatti'
},
{
  vehicle_name: 'M3 Competition',
  category: 'sport',
  manufacturer: 'BMW'
}
]

vehicles> use animal
switched to db animal
animal> show dbs
admin      40.00 KiB
config    96.00 KiB
local     40.00 KiB
vehicles  80.00 KiB
animal>
```

3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.



The screenshot shows a MongoDB terminal window with the following commands and output:

```
{
  bike_name: '1399SuperDukeR',
  category: '1358cc',
  manufacturer: 'KTM'
},
{
  bike_name: 'SuperMeteor650',
  category: '648cc',
  manufacturer: 'RoyalEnfield'
}
]

vehicles> db.four_wheelers.find({ performance: { $gt: 5 } }, { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 });
{
  vehicle_name: 'Q8RS', category: 'car', manufacturer: 'Audi' },
{
  vehicle_name: 'TharBoxx', category: 'car', manufacturer: 'Mahindra' },
{
  vehicle_name: 'Chiron',
  category: 'hypercar',
  manufacturer: 'Bugatti'
},
{
  vehicle_name: 'M3 Competition',
  category: 'sport',
  manufacturer: 'BMW'
}
]

vehicles> use animal
switched to db animal
animal> show dbs
admin      40.00 KiB
config    96.00 KiB
local     40.00 KiB
vehicles  80.00 KiB
animal> db.createCollection("wild_animals", { capped: true, size: 4096 });
{ ok: 1 }
animal> db.createCollection("domestic_animals");
{ ok: 1 }
animal>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

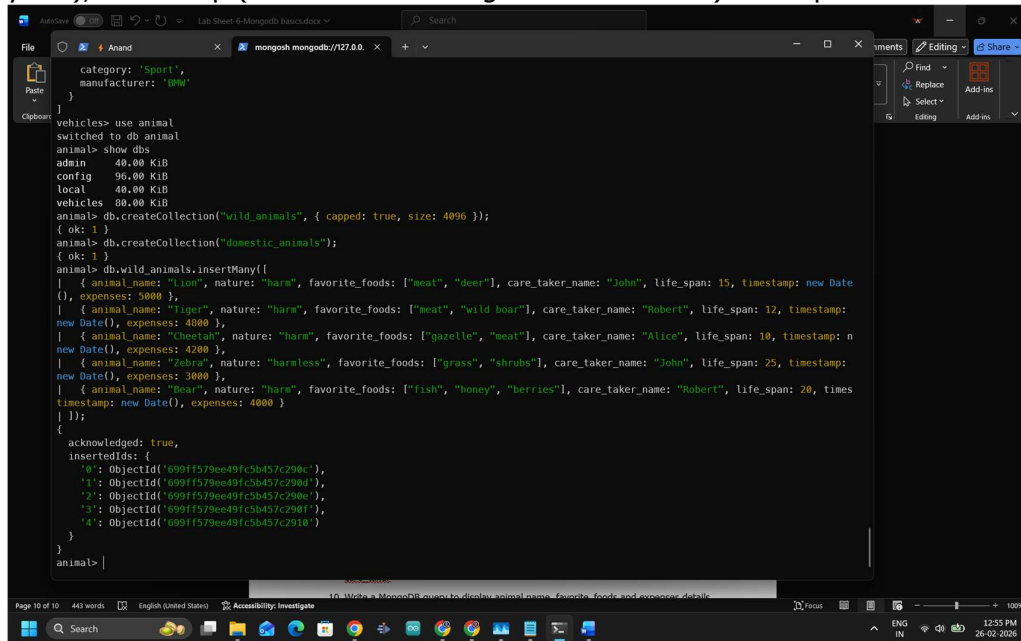
Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.



```
vehicles> use animal
switched to db animal
animal> show dbs
admin      48.00 KiB
config     96.00 KiB
local      48.00 KiB
vehicles    88.00 KiB
animal> db.createCollection("wild_animals", { capped: true, size: 4096 });
{ ok: 1 }
animal> db.createCollection("domestic_animals");
{ ok: 1 }
animal> db.wild_animals.insertMany([
  { animal_name: "Lion", nature: "harm", favorite_foods: ["meat", "deer"], care_taker_name: "John", life_span: 15, timestamp: new Date(), expenses: 5000 },
  { animal_name: "Tiger", nature: "harm", favorite_foods: ["meat", "wild boar"], care_taker_name: "Robert", life_span: 12, timestamp: new Date(), expenses: 4800 },
  { animal_name: "Gorilla", nature: "harm", favorite_foods: ["gazelle", "meat"], care_taker_name: "Alice", life_span: 10, timestamp: new Date(), expenses: 4200 },
  { animal_name: "Zebra", nature: "harmless", favorite_foods: ["grass", "shrubs"], care_taker_name: "John", life_span: 25, timestamp: new Date(), expenses: 3800 },
  { animal_name: "Bear", nature: "harm", favorite_foods: ["fish", "honey", "berries"], care_taker_name: "Robert", life_span: 20, timestamp: new Date(), expenses: 4000 }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699ff579ee49fc5b457c290c'),
    '1': ObjectId('699ff579ee49fc5b457c290d'),
    '2': ObjectId('699ff579ee49fc5b457c290e'),
    '3': ObjectId('699ff579ee49fc5b457c290f'),
    '4': ObjectId('699ff579ee49fc5b457c2910')
  }
}
animal>
```

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

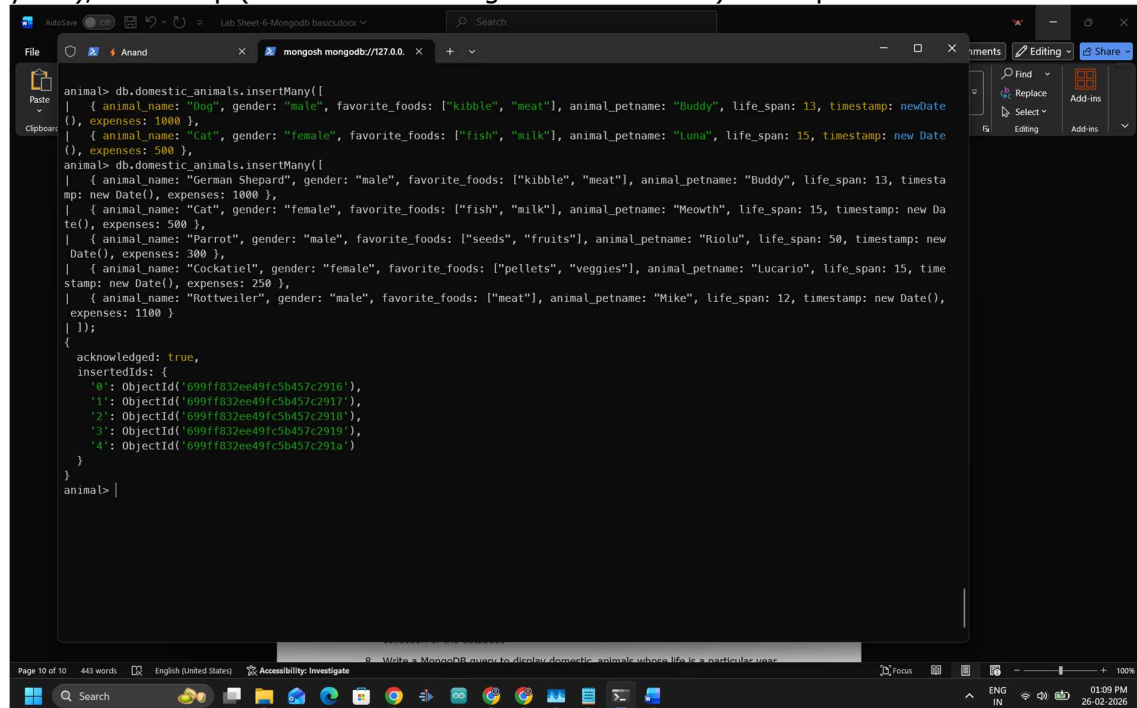
Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

years), timestamp (when the animal registered at the Zoo) and expenses.



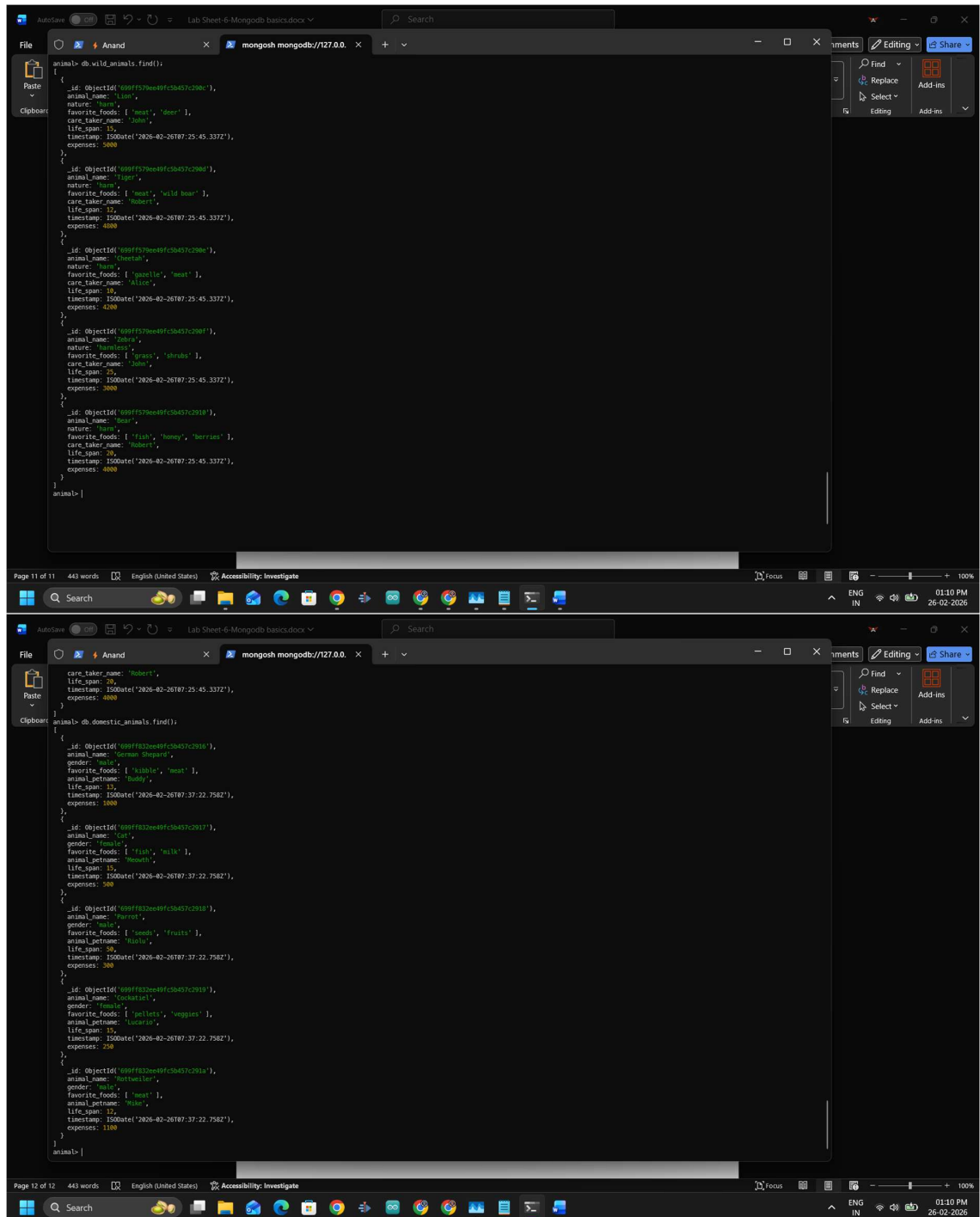
```
animal> db.domestic_animals.insertMany([
  { animal_name: "Dog", gender: "male", favorite_foods: ["kibble", "meat"], animal_petname: "Buddy", life_span: 13, timestamp: new Date(), expenses: 1000 },
  { animal_name: "Cat", gender: "female", favorite_foods: ["fish", "milk"], animal_petname: "Luna", life_span: 15, timestamp: new Date(), expenses: 500 },
  { animal_name: "German Shepard", gender: "male", favorite_foods: ["kibble", "meat"], animal_petname: "Buddy", life_span: 13, timestamp: new Date(), expenses: 1000 },
  { animal_name: "Cat", gender: "female", favorite_foods: ["fish", "milk"], animal_petname: "Meowth", life_span: 15, timestamp: new Date(), expenses: 500 },
  { animal_name: "Parrot", gender: "male", favorite_foods: ["seeds", "fruits"], animal_petname: "Riolu", life_span: 50, timestamp: new Date(), expenses: 300 },
  { animal_name: "Cockatiel", gender: "female", favorite_foods: ["pellets", "veggies"], animal_petname: "Lucario", life_span: 15, timestamp: new Date(), expenses: 250 },
  { animal_name: "Rottweiler", gender: "male", favorite_foods: ["meat"], animal_petname: "Mike", life_span: 12, timestamp: new Date(), expenses: 1100 }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699ff832ee49fc5b457c2916'),
    '1': ObjectId('699ff832ee49fc5b457c2917'),
    '2': ObjectId('699ff832ee49fc5b457c2918'),
    '3': ObjectId('699ff832ee49fc5b457c2919'),
    '4': ObjectId('699ff832ee49fc5b457c291a')
  }
}
animal> |
```

6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech
Faculty Name: Prof. S.Gopikrishnan
Student name: Anand Kumar V

Date: 26-Feb-26
School: SCOPE
Reg. no.: 23BCE8757



The image displays two screenshots of a MongoDB terminal window, likely using the mongosh shell. The top screenshot shows the results of a query on the 'wild_animals' collection, listing details for five animals including their IDs, names, natures, favorite foods, care takers, life spans, timestamps, and expenses. The bottom screenshot shows the results of a query on the 'domestic_animals' collection, listing details for five animals including their IDs, names, genders, favorite foods, care takers, life spans, timestamps, and expenses. Both screenshots show the terminal interface with a search bar, file explorer, and a sidebar with editing tools.

```
animals> db.wild_animals.find();
{
  "_id": ObjectId("599ff279e49fc5b457c296e"),
  "animal_name": "lion",
  "nature": "lion",
  "favorite_foods": [ "meat", "deer" ],
  "care_taker_name": "John",
  "life_span": 15,
  "timestamp": ISODate("2026-02-26T07:25:45.337Z"),
  "expenses": 5000
},
{
  "_id": ObjectId("599ff279e49fc5b457c296f"),
  "animal_name": "tiger",
  "nature": "lion",
  "favorite_foods": [ "meat", "wild bear" ],
  "care_taker_name": "Robert",
  "life_span": 12,
  "timestamp": ISODate("2026-02-26T07:25:45.337Z"),
  "expenses": 4000
},
{
  "_id": ObjectId("599ff279e49fc5b457c296e"),
  "animal_name": "cheetah",
  "nature": "lion",
  "favorite_foods": [ "meat", "meat" ],
  "care_taker_name": "Alice",
  "life_span": 10,
  "timestamp": ISODate("2026-02-26T07:25:45.337Z"),
  "expenses": 4200
},
{
  "_id": ObjectId("599ff279e49fc5b457c296f"),
  "animal_name": "zebra",
  "nature": "herbivore",
  "favorite_foods": [ "grass", "shrubs" ],
  "care_taker_name": "John",
  "life_span": 20,
  "timestamp": ISODate("2026-02-26T07:25:45.337Z"),
  "expenses": 3000
},
{
  "_id": ObjectId("599ff279e49fc5b457c296f"),
  "animal_name": "bear",
  "nature": "herb",
  "favorite_foods": [ "fish", "honey", "berries" ],
  "care_taker_name": "Robert",
  "life_span": 20,
  "timestamp": ISODate("2026-02-26T07:25:45.337Z"),
  "expenses": 4000
}
}
animals> |
```

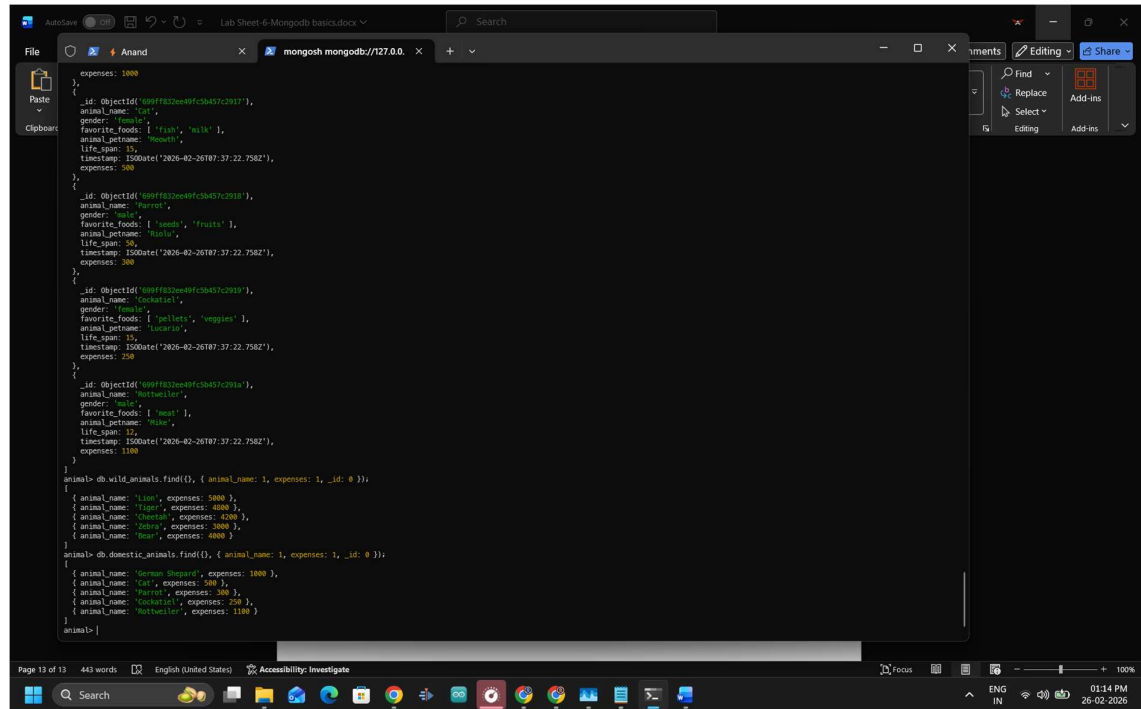
```
care_taker_name": "Robert",
life_span": 20,
timestamp": ISODate("2026-02-26T07:25:45.337Z"),
expenses": 4000
}
}
animals> db.domestic_animals.find();
{
  "_id": ObjectId("599ff832e49fc5b457c291a"),
  "animal_name": "German Shepherd",
  "gender": "male",
  "favorite_foods": [ "chicken", "meat" ],
  "animal_petname": "Buddy",
  "life_span": 15,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 1000
},
{
  "_id": ObjectId("599ff832e49fc5b457c2917"),
  "animal_name": "cat",
  "gender": "female",
  "favorite_foods": [ "fish", "milk" ],
  "animal_petname": "Mewmew",
  "life_span": 15,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 500
},
{
  "_id": ObjectId("599ff832e49fc5b457c291a"),
  "animal_name": "parrot",
  "gender": "male",
  "favorite_foods": [ "seeds", "fruits" ],
  "animal_petname": "Rishi",
  "life_span": 30,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 300
},
{
  "_id": ObjectId("599ff832e49fc5b457c2919"),
  "animal_name": "cockatiel",
  "gender": "female",
  "favorite_foods": [ "pellets", "vegetables" ],
  "animal_petname": "Lucario",
  "life_span": 15,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 250
},
{
  "_id": ObjectId("599ff832e49fc5b457c291a"),
  "animal_name": "robin",
  "gender": "male",
  "favorite_foods": [ "meat" ],
  "animal_petname": "Mike",
  "life_span": 12,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 1100
}
}
animals> |
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech
Faculty Name: Prof. S.Gopikrishnan
Student name: Anand Kumar V

Date: 26-Feb-26
School: SCOPE
Reg. no.: 23BCE8757

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database



The screenshot shows a MongoDB shell window with the following content:

```
expenses: 1000
{
  _id: ObjectId('699f832e49fc9b457c2817'),
  animal_name: 'Cat',
  gender: 'Male',
  favorite_foods: [ 'fish', 'meat' ],
  animal_petname: 'Mowth',
  life_span: 15,
  timestamp: ISODate('2026-02-26T07:37:22.758Z'),
  expenses: 500
},
{
  _id: ObjectId('699f832e49fc9b457c2818'),
  animal_name: 'Parrot',
  gender: 'Male',
  favorite_foods: [ 'seeds', 'fruits' ],
  animal_petname: 'Rulu',
  life_span: 30,
  timestamp: ISODate('2026-02-26T07:37:22.758Z'),
  expenses: 300
},
{
  _id: ObjectId('699f832e49fc9b457c2819'),
  animal_name: 'Cockatiel',
  gender: 'Female',
  favorite_foods: [ 'pellets', 'veggies' ],
  animal_petname: 'Lucaria',
  life_span: 15,
  timestamp: ISODate('2026-02-26T07:37:22.758Z'),
  expenses: 250
},
{
  _id: ObjectId('699f832e49fc9b457c281a'),
  animal_name: 'Rottweiler',
  gender: 'Male',
  favorite_foods: [ 'meat' ],
  animal_petname: 'Mue',
  life_span: 12,
  timestamp: ISODate('2026-02-26T07:37:22.758Z'),
  expenses: 1100
}
}

animal> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 });
{
  animal_name: 'Lion', expenses: 5000 },
{
  animal_name: 'Tiger', expenses: 4000 },
{
  animal_name: 'Cheetah', expenses: 4200 },
{
  animal_name: 'Zebra', expenses: 3000 },
{
  animal_name: 'Gorilla', expenses: 4000 }
}

animal> db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 });
{
  animal_name: 'German Shepherd', expenses: 1000 },
{
  animal_name: 'Cat', expenses: 500 },
{
  animal_name: 'Parrot', expenses: 300 },
{
  animal_name: 'Cockatiel', expenses: 250 },
{
  animal_name: 'Rottweiler', expenses: 1100 }
}

animal> |
```

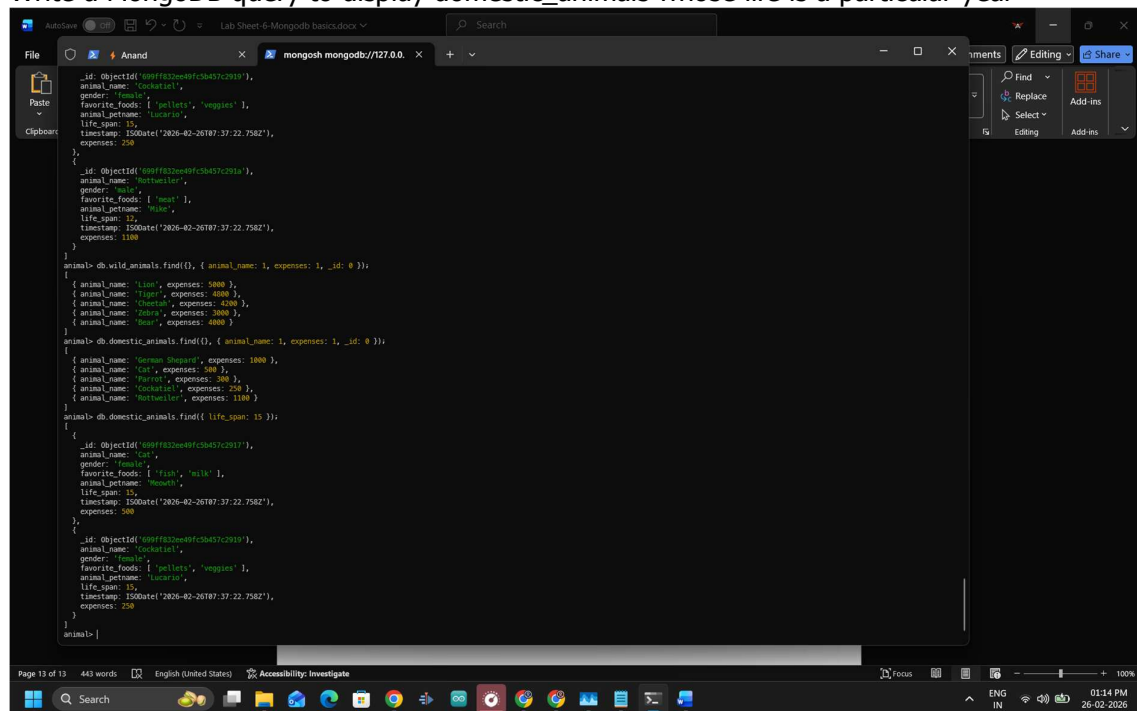
The bottom status bar of the application shows: Page 13 of 13, 443 words, English (United States), Accessibility: Investigate, ENG IN, 01:14 PM, 26-02-2026.

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech
Faculty Name: Prof. S.Gopikrishnan
Student name: Anand Kumar V

Date: 26-Feb-26
School: SCOPE
Reg. no.: 23BCE8757

8. Write a MongoDB query to display domestic_animals whose life is a particular year



```

{
  "_id": ObjectId("60ff8220e49fc9a437c2819"),
  "animal_name": "cockatiel",
  "gender": "female",
  "favorite_foods": [ "pellets", "veggies" ],
  "animal_getname": "lucaris",
  "life_span": 15,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 250
},
{
  "_id": ObjectId("60ff8220e49fc9a437c281a"),
  "animal_name": "rottwheeler",
  "gender": "male",
  "favorite_foods": [ "meat" ],
  "animal_getname": "Mike",
  "life_span": 15,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 1100
}
}

animal> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 });
{
  "animal_name": "lion", "expenses": 5000 },
{
  "animal_name": "tiger", "expenses": 4000 },
{
  "animal_name": "cheetah", "expenses": 4000 },
{
  "animal_name": "leopard", "expenses": 3000 },
{
  "animal_name": "bear", "expenses": 4000 }
}

animal> db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 });
{
  "animal_name": "German Shepherd", "expenses": 1000 },
{
  "animal_name": "cat", "expenses": 500 },
{
  "animal_name": "Parrot", "expenses": 300 },
{
  "animal_name": "cockatiel", "expenses": 250 },
{
  "animal_name": "rottwheeler", "expenses": 1100 }
}

animal> db.domestic_animals.find({ life_span: 15 });
{
  "_id": ObjectId("60ff8220e49fc9a437c2817"),
  "animal_name": "cat",
  "gender": "female",
  "favorite_foods": [ "fish", "meat" ],
  "animal_getname": "Mowath",
  "life_span": 15,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 500
},
{
  "_id": ObjectId("60ff8220e49fc9a437c2819"),
  "animal_name": "cockatiel",
  "gender": "female",
  "favorite_foods": [ "pellets", "veggies" ],
  "animal_getname": "lucaris",
  "life_span": 15,
  "timestamp": ISODate("2026-02-26T07:37:22.758Z"),
  "expenses": 250
}
}

animal>

```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

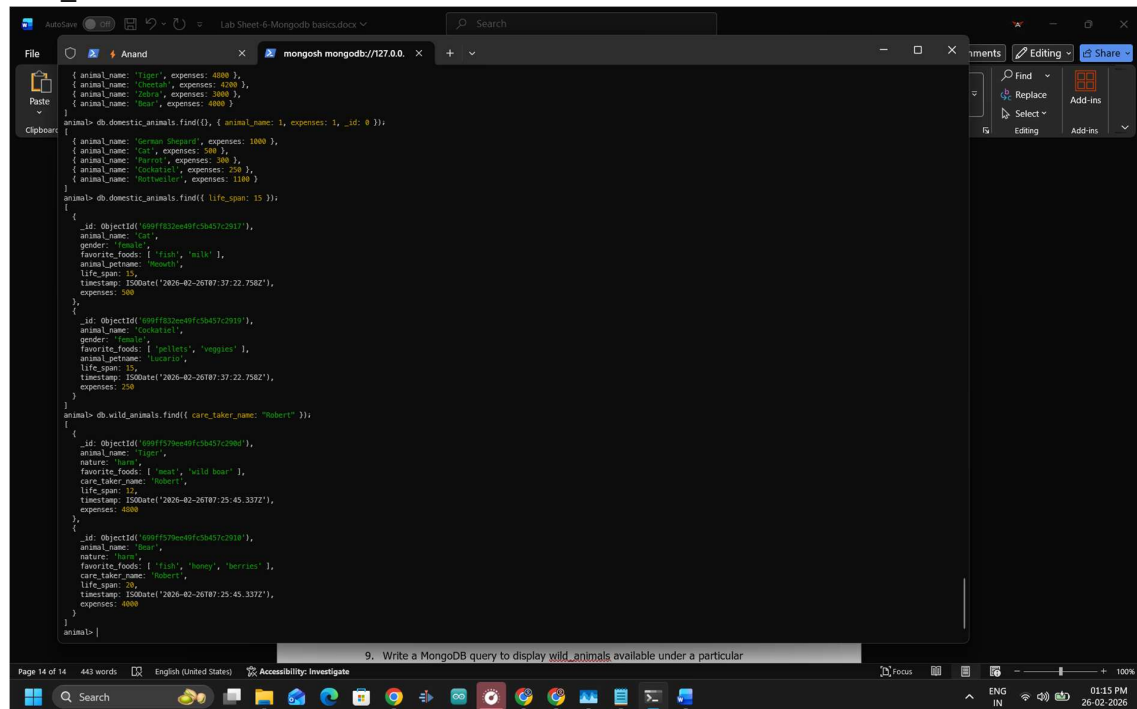
Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

9. Write a MongoDB query to display wild_animals available under a particular care_taker



The screenshot shows a MongoDB shell window with the following content:

```

{ animal_name: "tiger", expenses: 4000 },
{ animal_name: "cheetah", expenses: 4200 },
{ animal_name: "leopard", expenses: 3800 },
{ animal_name: "bear", expenses: 4500 }
}
animal> db.domestic_animals.find({ animal_name: 1, expenses: 1, _id: 0 });
{
  animal_name: "German Shepherd", expenses: 1000 },
{ animal_name: "cat", expenses: 500 },
{ animal_name: "Parrot", expenses: 300 },
{ animal_name: "chihuahua", expenses: 250 },
{ animal_name: "rottweiler", expenses: 1100 }
}
animal> db.domestic_animals.find({ life_span: 15 });
{
  _id: ObjectId("699f832ae0fc5b457c2017"),
  animal_name: "cat",
  gender: "female",
  favorite_foods: [ "fish", "milk" ],
  animal_petname: "Mewch",
  life_span: 15,
  timestamp: ISODate("2026-02-26T07:37:22.758Z"),
  expenses: 500
},
{
  _id: ObjectId("699f832ae0fc5b457c2019"),
  animal_name: "chihuahua",
  gender: "female",
  favorite_foods: [ "pellets", "veggies" ],
  animal_petname: "Kuturo",
  life_span: 15,
  timestamp: ISODate("2026-02-26T07:37:22.758Z"),
  expenses: 250
}
}
animal> db.wild_animals.find({ care_taker_name: "Robert" });
{
  _id: ObjectId("699f832ae0fc5b457c200a"),
  animal_name: "lion",
  nature: "harm",
  favorite_foods: [ "meat", "wild bear" ],
  care_taker_name: "Robert",
  life_span: 15,
  timestamp: ISODate("2026-02-26T07:25:45.337Z"),
  expenses: 4000
},
{
  _id: ObjectId("699f832ae0fc5b457c201a"),
  animal_name: "bear",
  nature: "harm",
  favorite_foods: [ "fish", "honey", "berries" ],
  care_taker_name: "Robert",
  life_span: 20,
  timestamp: ISODate("2026-02-26T07:25:45.337Z"),
  expenses: 4000
}
}
animal> |

```

9. Write a MongoDB query to display wild_animals available under a particular

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 26-Feb-26

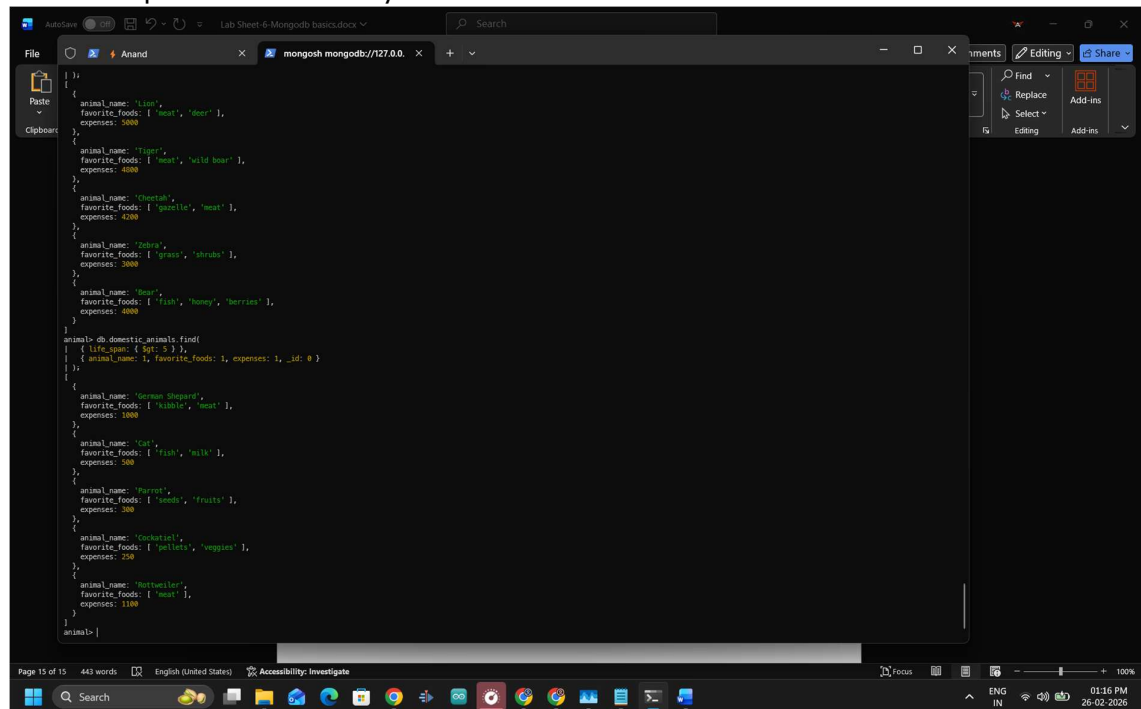
Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Anand Kumar V

Reg. no.: 23BCE8757

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.



The screenshot shows a MongoDB shell window with a list of animals and a query to filter by lifespan. The list of animals is as follows:

```
{
  animal_name: 'Lion',
  favorite_foods: [ 'meat', 'deer' ],
  expenses: 5000
},
{
  animal_name: 'Tiger',
  favorite_foods: [ 'meat', 'wild boar' ],
  expenses: 4000
},
{
  animal_name: 'Gorilla',
  favorite_foods: [ 'banana', 'meat' ],
  expenses: 4200
},
{
  animal_name: 'Zebra',
  favorite_foods: [ 'grass', 'shrubs' ],
  expenses: 3000
},
{
  animal_name: 'Bear',
  favorite_foods: [ 'fish', 'honey', 'berries' ],
  expenses: 4000
}
}
```

The query executed is:

```
animal> db.domestic_animals.find(
  { life_span: { $gt: 5 } },
  { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
);
```

The results of the query are:

```
{
  animal_name: 'German Shepherd',
  favorite_foods: [ 'apple', 'meat' ],
  expenses: 1000
},
{
  animal_name: 'Cat',
  favorite_foods: [ 'fish', 'milk' ],
  expenses: 500
},
{
  animal_name: 'Parrot',
  favorite_foods: [ 'nuts', 'fruits' ],
  expenses: 300
},
{
  animal_name: 'Chickadee',
  favorite_foods: [ 'pollen', 'vegetables' ],
  expenses: 250
},
{
  animal_name: 'Robin',
  favorite_foods: [ 'meat' ],
  expenses: 1100
}
}
```